

1. Project Overview

N100 Financial Intelligence Platform is a financial analytics system built for analysing 92 Nifty 100 companies.

The platform provides:

- Financial KPI analysis
- Investment screening
- Peer comparison
- Sector analysis
- Company profile analytics
- Automated PDF tearsheets

The system uses ETL pipelines, SQLite database, FastAPI backend and Streamlit dashboard.

2. System Architecture

The platform consists of four major layers:

1. Data Layer:

Stores financial statements, ratios, market data and company information.

2. Analytics Layer:

Calculates KPIs such as ROE, ROCE, CAGR, Debt Equity ratio and quality scores.

3. API Layer:

FastAPI exposes financial data through REST endpoints.

4. Dashboard Layer:

Streamlit provides analyst-friendly visualization.

3. Setting Up The Project

Steps:

1. Activate virtual environment.
2. Install requirements.
3. Load financial datasets.
4. Run ETL pipeline.
5. Start API server.
6. Start Streamlit dashboard.

4. Using The Screener

The screener allows analysts to filter companies based on:

- ROE
- ROCE
- Debt levels
- Quality score
- Growth metrics

Users can select filters and export results as CSV.

5. Dashboard Navigation

The dashboard contains:

Company Explorer:

View individual company information.

Screener:

Filter investment opportunities.

Peer Analysis:

Compare companies against industry peers.

Sector Dashboard:

Analyse sector-level trends.

6. Company Profile Screen

The company profile displays:

- Latest financial metrics
- Profit and loss history
- Balance sheet information
- Cash flow trends
- Financial ratios
- Peer comparison

7. PDF Tearsheet Generation

The platform generates automated analyst tearsheets.

Each tearsheet contains:

- Company overview
- Financial charts
- KPI summary
- Quality indicators
- Investment insights

Generated files are stored inside:

reports/tearsheets/

8. API Usage

FastAPI provides REST endpoints.

Example:

GET /api/v1/companies/TCS

Returns company information.

GET /api/v1/screener

Returns screening results.

Health check:

GET /api/v1/health

9. Troubleshooting

Common issues:

Database not found:
Check database path in settings.

API connection error:
Ensure FastAPI server is running.

Dashboard loading error:
Verify API endpoint availability.

10. Analyst Workflow

Recommended workflow:

1. Open dashboard.
2. Select sector/company.
3. Analyse KPIs.
4. Compare peers.
5. Run screener.
6. Export results.
7. Generate tearsheet.